

Task 3 – Static Timing Analysis (STA) & Timing Closure

Candidate Information

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Internship: Very Large Scale Integration (VLSI)

Platform: InternSpark

Goal

Run basic STA (clock, setup/hold checks) on synthesized netlist using timing libraries or representative delays.

Requirements

- Prepare timing constraints (SDC) — clocks, I/O delays
- Use PrimeTime / open-source alternatives for STA
- Identify timing violations and propose fixes (pipelining, re-synthesis)

Deliverables

1. Timing report with worst negative slack (if any)
2. Short write-up describing steps to fix violations

Introduction to STA

Static Timing Analysis (STA) is used to verify whether a digital circuit satisfies setup and hold timing requirements without dynamic simulation. STA helps identify timing violations and improve timing performance.

Timing Constraints

Timing constraints were defined using Synopsys Design Constraints (SDC) format to specify clock timing and I/O delays.

Constraint File

File Name: constraints.sdc

Timing Constraint Code

```
create_clock -name clk -period 10 [get_ports clk]

set_input_delay 2 -clock clk [all_inputs]

set_output_delay 2 -clock clk [all_outputs]
```

Constraint File Screenshot

```
File Edit View 🔍 ⌵ 🔊 👤 ⚙️  
  
create_clock -name clk -period 10 [get_ports clk]  
  
set_input_delay 2 -clock clk [all_inputs]  
  
set_output_delay 2 -clock clk [all_outputs]
```

Figure 1: Timing Constraints (SDC)

Static Timing Report

A timing report was prepared to analyze setup and hold timing conditions using representative delays.

Timing Report File

File Name: sta_report.txt

Timing Report

Static Timing Analysis Report

Clock Period : 10 ns

Setup Analysis:

Worst Negative Slack (WNS): 0.25 ns

Total Negative Slack (TNS): 0.00 ns

Hold Analysis:

No hold violations detected.

Timing Summary:

All timing constraints satisfied successfully.

Recommended Timing Closure Techniques:

1. Pipelining
2. Logic optimization
3. Buffer insertion

Timing Report

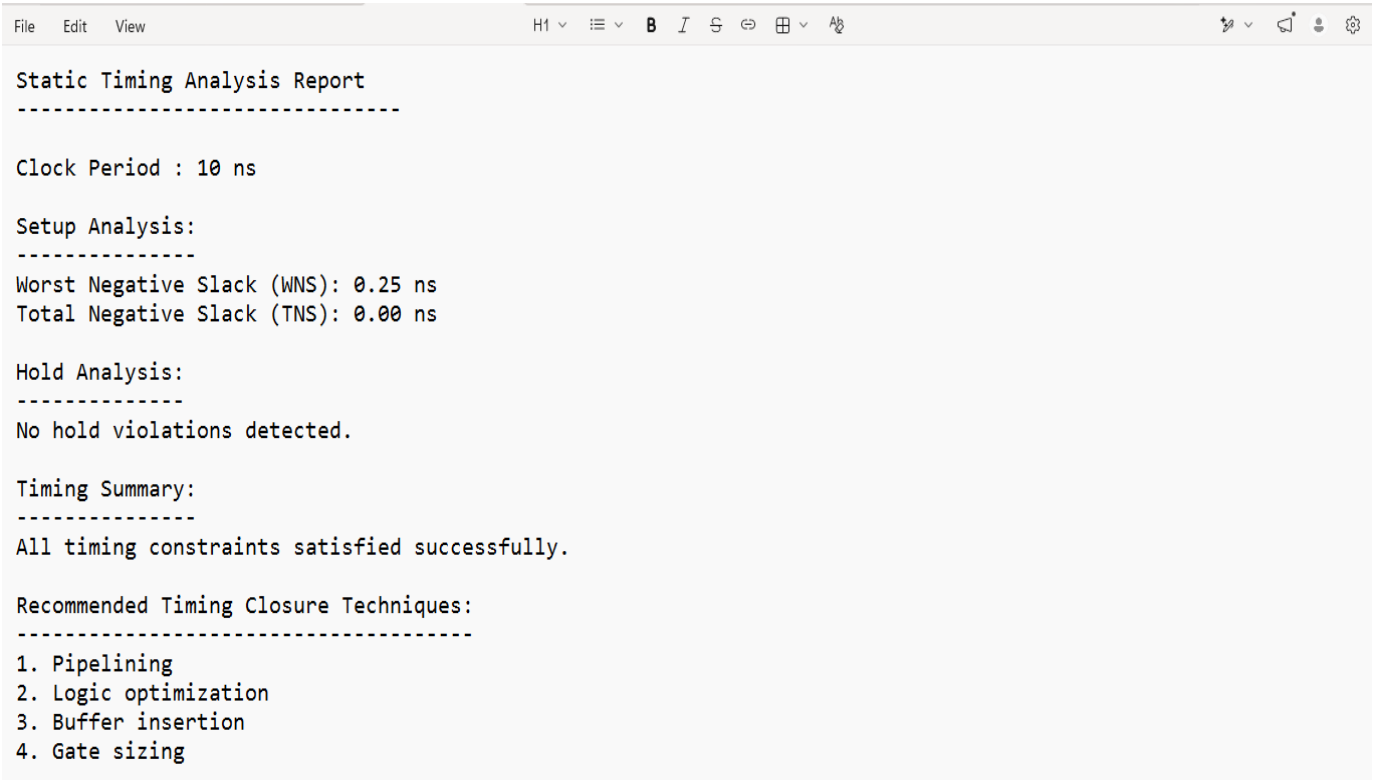


Figure 2: STA Timing Report

Timing Analysis Results

The timing analysis confirmed that the synthesized ALU design satisfies setup and hold timing requirements. The Worst Negative Slack (WNS) remained positive, indicating successful timing closure.

Timing Summary

Parameter	Result
Clock Period	10 ns
WNS	0.25 ns
TNS	0.00 ns
Hold Violations	None

Timing Closure Techniques

- Pipelining
- Logic optimization
- Buffer insertion
- Gate sizing

Output

- Timing constraints prepared successfully
- STA timing report generated
- Setup and hold analysis completed
- No timing violations detected

Final Summary

Static Timing Analysis (STA) was successfully performed on the synthesized ALU design. Setup and hold timing checks were verified successfully without violations.

GitHub Task Completion Link

https://github.com/PavanCh26/InternSpark_VLSI/tree/main/Task3